RENEWABLE ENERGY PERFORMANCE ANALYTICS

Data Analyst Project — Python + MySQL + Power BI + Streamlit

Final simplified project structure for a Data Analyst portfolio

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1. Project Overview

A renewable energy company operates solar, wind and hydro power plants across different cities and regions. The company wants to understand energy generation, plant efficiency, capacity utilization, downtime, weather impact and operating costs.

The objective is to analyze renewable energy performance, identify important business patterns and present actionable insights using **Python, MySQL, Power BI and a simple Streamlit data explorer.**

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2. Project Scope

- **Python** — data cleaning, EDA, analysis and derived business metrics.
- **MySQL** — database creation, data storage and analytical SQL queries.
- **Power BI** — main dashboard and business visualization.
- **Streamlit** — simple interactive energy/data explorer only.
- No Machine Learning or prediction model.
- No separate Streamlit dashboard because the main dashboard is created in Power BI.
- No SQLite; MySQL is the only database used.

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3. Business Objectives

1. Analyze total renewable energy generation.

2. Compare Solar, Wind and Hydro energy performance.
3. Identify high-performing and low-performing plants.
4. Analyze capacity utilization and plant efficiency.
5. Analyze the effect of weather conditions on energy generation.
6. Analyze downtime and maintenance performance.
7. Analyze operating and maintenance costs.
8. Analyze revenue generated by different energy sources.
9. Build an interactive Power BI dashboard.
10. Provide a simple Streamlit interface to explore energy records.

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4. Dataset

Use a renewable-energy dataset concept with **50,000+ energy generation and operational records.**

Category	**Columns**	**Purpose**
Energy Plant	Plant_ID, Plant_Name, Energy_Type, Region, City	Plant and location analysis
Generation	Record_ID, Date, Hour, Energy_Generated_MWh	Energy generation tracking
Capacity	Installed_Capacity_MW, Capacity_Utilization	Capacity analysis
Weather	Temperature_C, Wind_Speed_kmh, Solar_Irradiance, Rainfall_mm	Weather impact analysis
Operations	Downtime_Hours, Maintenance_Hours, Plant_Status	Operational analysis
Cost	Operating_Cost, Maintenance_Cost, Revenue	Cost and revenue analysis
Performance	Efficiency_Percent, Availability_Percent	Performance analysis

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5. Python

5.1 Libraries

- Pandas

- NumPy
- Matplotlib
- Seaborn

5.2 Data Cleaning

- Load raw renewable-energy CSV.
- Check rows, columns and data types.
- Handle missing values.
- Remove duplicate records where required.
- Convert date and time columns to appropriate formats.
- Validate numeric columns.
- Standardize categorical values.
- Check unrealistic or invalid values.
- Save cleaned_energy_data.csv.

5.3 EDA

- Total energy generated.
- Energy generation trend.
- Generation by Solar, Wind and Hydro.
- Generation by plant.
- Generation by region.
- Capacity utilization analysis.
- Plant efficiency comparison.
- Weather versus energy generation.
- Downtime versus energy generation.
- Maintenance cost analysis.
- Operating cost analysis.
- Revenue analysis.
- Plant availability analysis.

5.4 Derived Metrics

- Total Energy Generated (MWh).

- Average Daily Generation.
- Capacity Utilization Rate.
- Plant Efficiency Rate.
- Plant Availability Rate.
- Downtime Rate.
- Cost per MWh.
- Revenue per MWh.
- Maintenance Cost per MWh.
- Plant Performance Score.

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6. MySQL

6.1 Database

Create a MySQL database named:

renewable_energy_analytics

6.2 Tables

- Plants
- Energy_Generation
- Weather_Data
- Maintenance_Records

6.3 SQL Concepts

- CREATE DATABASE and CREATE TABLE.
- INSERT and SELECT.
- WHERE, GROUP BY, HAVING and ORDER BY.
- INNER JOIN and LEFT JOIN.
- CASE statements.
- CTEs.
- Window Functions such as RANK() and DENSE_RANK().
- Views.
- COUNT, SUM, AVG, MIN and MAX.

6.4 Business SQL Questions

- Which plants generate the highest amount of renewable energy?
- Which energy type has the highest average generation?
- Which plants have the highest capacity utilization?
- Which regions generate the most energy?
- Which plants have the highest downtime?
- How does weather affect energy generation?
- Which plants have the lowest cost per MWh?
- What are the top 10 plants by performance?
- Which months have the highest energy generation?
- Which plants have high capacity but relatively low generation?

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7. Power BI — Main Dashboard

Power BI is the main visualization and dashboard tool for this project. The Streamlit application will not duplicate this dashboard.

Page 1 — Executive Overview

- Total Energy Generated
- Installed Capacity
- Average Efficiency
- Capacity Utilization
- Total Revenue
- Total Operating Cost
- Energy Generation Trend
- Generation by Energy Type

Page 2 — Plant Performance

- Energy generation by plant
- Average efficiency by plant
- Capacity utilization
- Plant availability

- High-performing plants
- Low-performing plants
- Plant-wise cost per MWh

Page 3 — Renewable Energy Type Analytics

- Solar vs Wind vs Hydro generation
- Average efficiency by energy type
- Installed capacity comparison
- Revenue comparison
- Cost per MWh comparison
- Monthly generation trends

Page 4 — Weather & Operations

- Solar irradiance vs solar generation
- Wind speed vs wind generation
- Rainfall vs hydro generation
- Downtime vs energy generation
- Maintenance hours vs efficiency
- Plant availability

Page 5 — Cost & Revenue Analytics

- Operating cost by plant
- Maintenance cost by plant
- Cost per MWh
- Revenue by energy type
- Revenue vs energy generation
- Monthly cost and revenue trends

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8. Streamlit — Simple Data Explorer

Streamlit is used only as a small Python application to demonstrate interactive data exploration. It is not a second Power BI dashboard.

8.1 What the Streamlit App Will Do

- Read the cleaned energy dataset.
- Display a simple energy data table.
- Allow the user to filter energy records.
- Show basic summary values for filtered data.
- Display selected plant details.
- Allow filtered data to be downloaded as CSV.

8.2 Simple Filters

- Energy Type
- Plant
- Region
- Date Range
- Plant Status

8.3 Basic Summary

- Number of filtered records.
- Total energy generated.
- Average efficiency.
- Total operating cost.
- Average capacity utilization.

8.4 Energy Record Details

- Plant ID
- Plant Name
- Energy Type
- Region
- Date
- Energy Generated
- Installed Capacity
- Efficiency
- Downtime
- Operating Cost

8.5 Streamlit Command

Install packages:

pip install streamlit pandas

Run the application from the project root:

streamlit run streamlit_app/app.py

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9. End-to-End Workflow

1. Load raw_energy_data.csv in Python.
2. Clean and validate the data.
3. Perform EDA and calculate business metrics.
4. Save cleaned_energy_data.csv.
5. Create the renewable_energy_analytics database in MySQL.
6. Create Plants, Energy_Generation, Weather_Data and Maintenance_Records tables.
7. Insert cleaned data into MySQL.
8. Write analytical MySQL queries, CTEs, window functions and views.
9. Connect the analytical data to Power BI.
10. Build the main Power BI dashboard.
11. Create the simple Streamlit data explorer.
12. Validate important numbers across Python, MySQL, Power BI and Streamlit.
13. Prepare final business insights.

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10. Final Deliverables

- data/raw_energy_data.csv
- data/cleaned_energy_data.csv
- python/Data_Cleaning.ipynb
- python/EDA.ipynb
- python/Performance_Analysis.ipynb
- mysql/schema.sql

- mysql/queries.sql
- mysql/views.sql
- powerbi/Renewable_Energy_Performance_Analytics.pbix
- streamlit_app/app.py
- README.md

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11. Project Folder Structure

Renewable_Energy_Performance_Analytics_Project/

```
│
├── data/
│   ├── raw_energy_data.csv
│   └── cleaned_energy_data.csv
│
├── python/
│   ├── Data_Cleaning.ipynb
│   ├── EDA.ipynb
│   └── Performance_Analysis.ipynb
│
├── mysql/
│   ├── schema.sql
│   ├── queries.sql
│   └── views.sql
│
├── powerbi/
│   └── Renewable_Energy_Performance_Analytics.pbix
│
├── streamlit_app/
│   └── app.py
```

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│
└── README.md
```

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12. Expected Business Insights

- Identify plants with higher and lower observed energy generation.
- Compare Solar, Wind and Hydro energy performance.
- Identify plants with higher capacity utilization.
- Understand the relationship between weather conditions and energy generation.
- Identify plants with higher downtime or maintenance requirements.
- Compare operating and maintenance costs.
- Identify periods with unusual energy generation patterns.
- Identify differences in efficiency across plants and energy types.
- Analyze cost and revenue patterns.
- Provide evidence-based operational observations from the analyzed data.

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13. Data Analyst Viva Questions

1. Why did you choose this renewable energy project?
2. How did you clean the dataset?
3. How did you calculate capacity utilization?
4. How did you calculate cost per MWh?
5. How did you evaluate plant performance?
6. Where did you use MySQL joins?
7. Where did you use a CTE?
8. Where did you use a window function?
9. Why did you use Power BI?
10. Why did you use Streamlit if Power BI already has the dashboard?
11. How did weather variables relate to energy generation?
12. How did you validate your results?

13. What business insights did you find from the data?
14. Which renewable energy type generated the most energy?
15. Which factors affected plant performance?

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14. Final Technology Stack

Technology	Purpose
Python	Cleaning, EDA and analysis
MySQL	Database and SQL analysis
Power BI	Main dashboard and visualization
Streamlit	Simple interactive energy/data explorer

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15. Scope Note

This project is based on the **Renewable Energy Performance Analytics** scenario. The Data Analyst version focuses on **Python, MySQL and Power BI**, with Streamlit retained only as a simple data-exploration application.

The main dashboard remains in Power BI.

Python + MySQL + Power BI + Simple Streamlit Data Explorer